Revenue story at a glance

This dataset brings together **10,388 sales records**, **497 customers**, **28 products**, and **14 cities**, giving a clean view of how revenue is built across time, products, customers, and locations. The clearest headline is that **Pune is the top-performing city by revenue**, and its lead comes more from **how often customers buy** than from **larger order values**. Promotion modeling adds a second important finding: **discounts help revenue only when customer demand responds strongly enough**.

The dataset is clean and ready for business analysis

The four files connect neatly, which makes the results easier to trust. There are no missing cells in any table, and the relationships between sales, products, customers, and cities all line up without gaps.

table	rows	cols
city	14	5
customers	497	3
products	28	3
sales	10388	6

table	missing_cells
city	0
customers	0
products	0
sales	0

check	count
sales.product_id missing in products	0
sales.customer_id missing in customers	0

check	count
customers.city_id missing in city	0

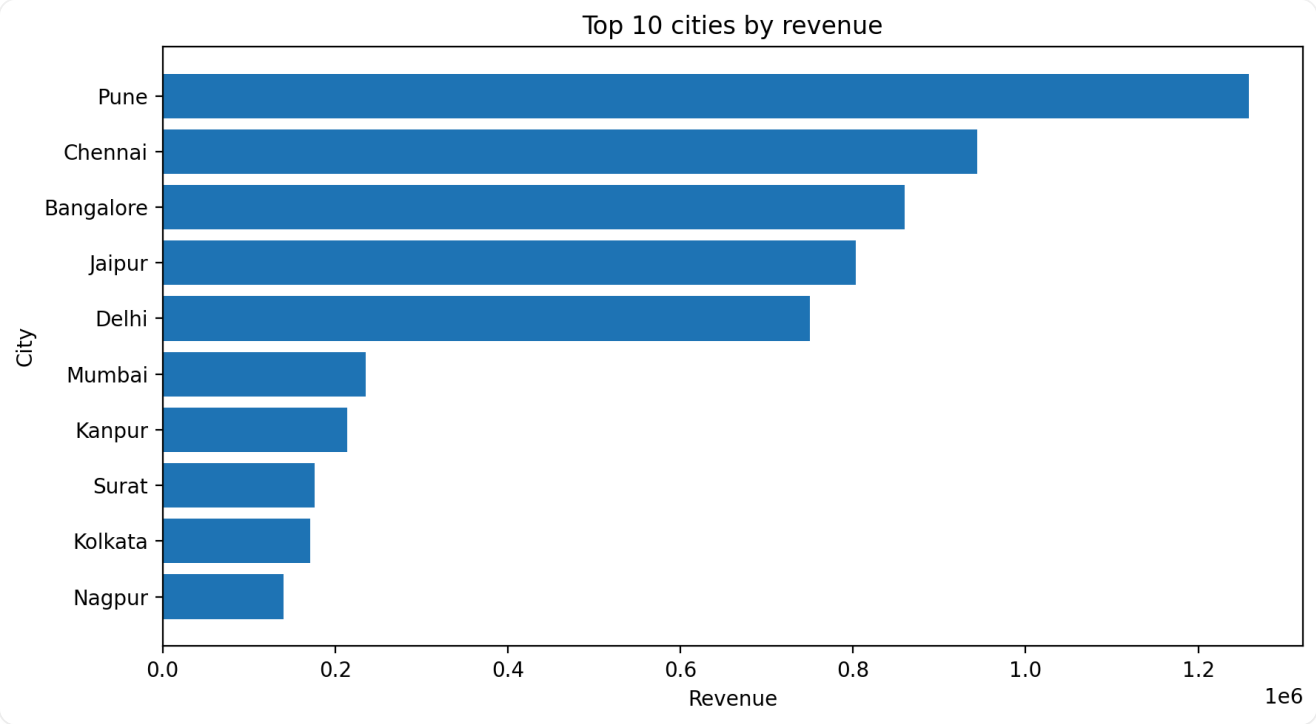
This structure supports practical questions around revenue growth, city performance, product demand, customer loyalty, and the likely effect of promotions.

Pune leads all cities on revenue

Pune generated the highest total revenue at **1,258,290**, ahead of Chennai and Bangalore.

city_id	city_name	total
3	Pune	1258290
2	Chennai	944120
1	Bangalore	860110
4	Jaipur	803450
5	Delhi	750420
6	Mumbai	235000
12	Kanpur	213550
10	Surat	176540
9	Kolkata	171460
13	Nagpur	140050

Download



The chart shows a clear gap between Pune and the rest of the field, with Chennai as the nearest follower.

Pune's edge comes from repeat buying, not bigger baskets

Pune's revenue lead is mainly explained by **higher order volume** and **strong repeat purchasing**. Its average order value is almost the same as Chennai's, so the city is not winning because people spend more each time.

city_name	revenue	orders	customers	avg_order_valu
Pune	1258290	2135	52	589.362997658
Chennai	944120	1601	42	589.706433479
Bangalore	860110	1464	39	587.506830601
Jaipur	803450	1377	69	583.478576615
Delhi	750420	1305	68	575.034482758
Mumbai	235000	423	27	555.555555555
Kanpur	213550	372	35	574.059139784
Surat	176540	297	27	594.410774410

A simpler comparison makes the pattern even clearer:

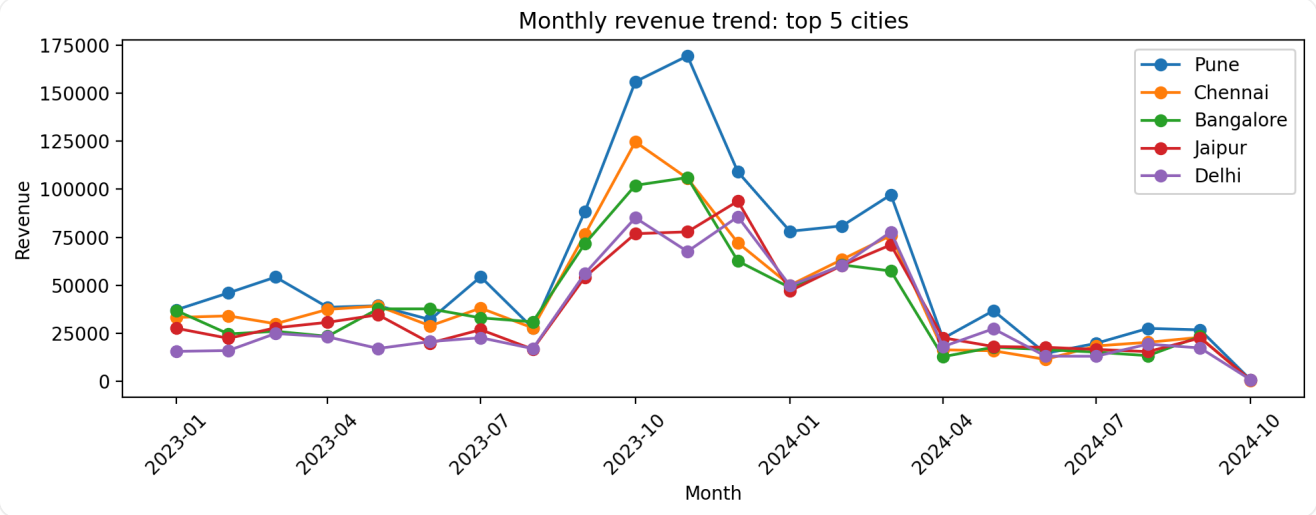
segment	revenue	orders	customers	avg_order_value
Pune	1258290	2135	52	589.3629976580
Chennai	944120	1601	42	589.7064334790
Overall	6070190	10388	497	584.3463611859

Pune has only about a tenth of customers overall, but contributes about a fifth of all orders and revenue.

metric	Pune
Customer share	0.10462776659959759
Order share	0.2055256064690027
Revenue share	0.20729005187646515

That makes Pune's story straightforward: **customers there come back more often**, and that repeat behavior lifts revenue.

Download



This city trend chart supports the same story by showing Pune consistently among the strongest monthly contributors.

Pune wins with the same winning products, just more often

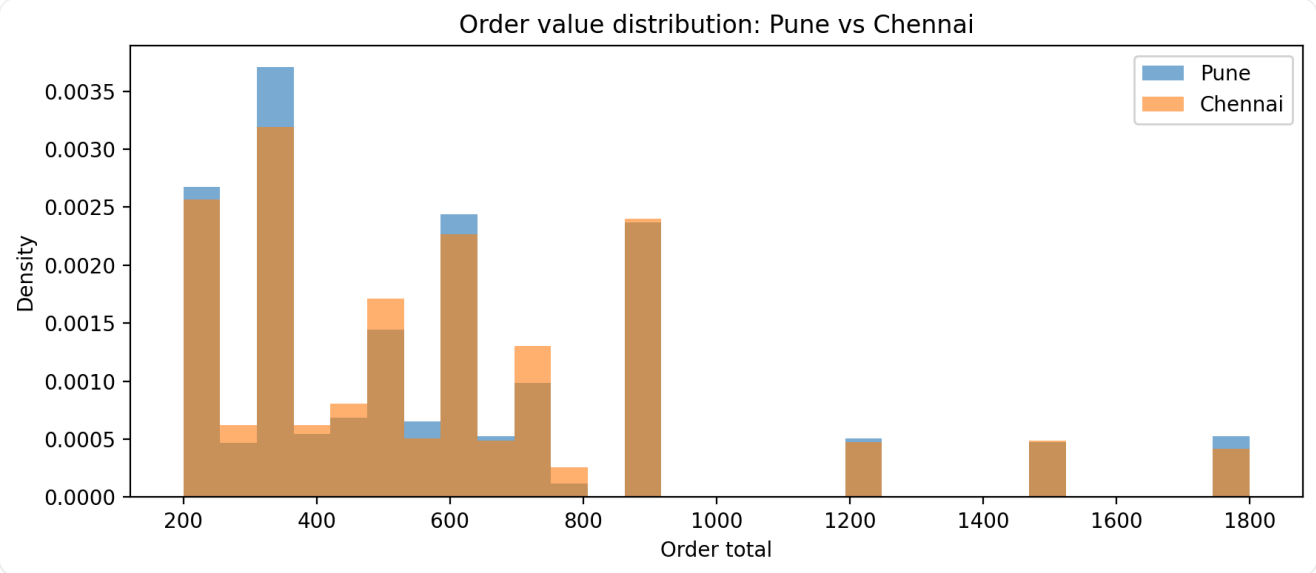
Pune's best-selling products are not unusual outliers. They are largely the same products that perform well across the full business, which suggests Pune's lead is not being driven by one-off demand.

product_name	total
Cold Brew Coffee Pack (6 Bottles)	233100
Coffee Beans (500g)	144000
Coffee Gift Hamper	111600
Ground Espresso Coffee (250g)	88900
Specialty Coffee Subscription	84000
Tote Bag with Coffee Design	76000
French Press Coffee Set	72000
Instant Coffee Powder (100g)	61250
Vanilla Coffee Syrup (250ml)	53440
Flavored Coffee Pods (Pack of 10)	47250

product_name	total
Cold Brew Coffee Pack (6 Bottles)	1193400
Coffee Beans (500g)	730800
Coffee Gift Hamper	486000
Ground Espresso Coffee (250g)	444850
Tote Bag with Coffee Design	388000
Specialty Coffee Subscription	387000
French Press Coffee Set	308400
Instant Coffee Powder (100g)	306500
Vanilla Coffee Syrup (250ml)	243840
Flavored Coffee Pods (Pack of 10)	221250

So Pune's advantage looks less like a special product mix and more like **strong adoption of the same core bestsellers**.

Download



The order value comparison reinforces that Pune is not pulling ahead because of bigger order sizes.

Promotions can raise revenue, but only if demand reacts enough

The overall business baseline is:

- **Revenue:** 6,070,190
- **Orders:** 10,388
- **Average order value:** 584.35

The promotion modeling tested discounts from **5% to 30%** under different levels of customer responsiveness. The message is simple:

- If customers react strongly to lower prices, revenue can grow.
- If they react only a little, discounts mostly reduce revenue.

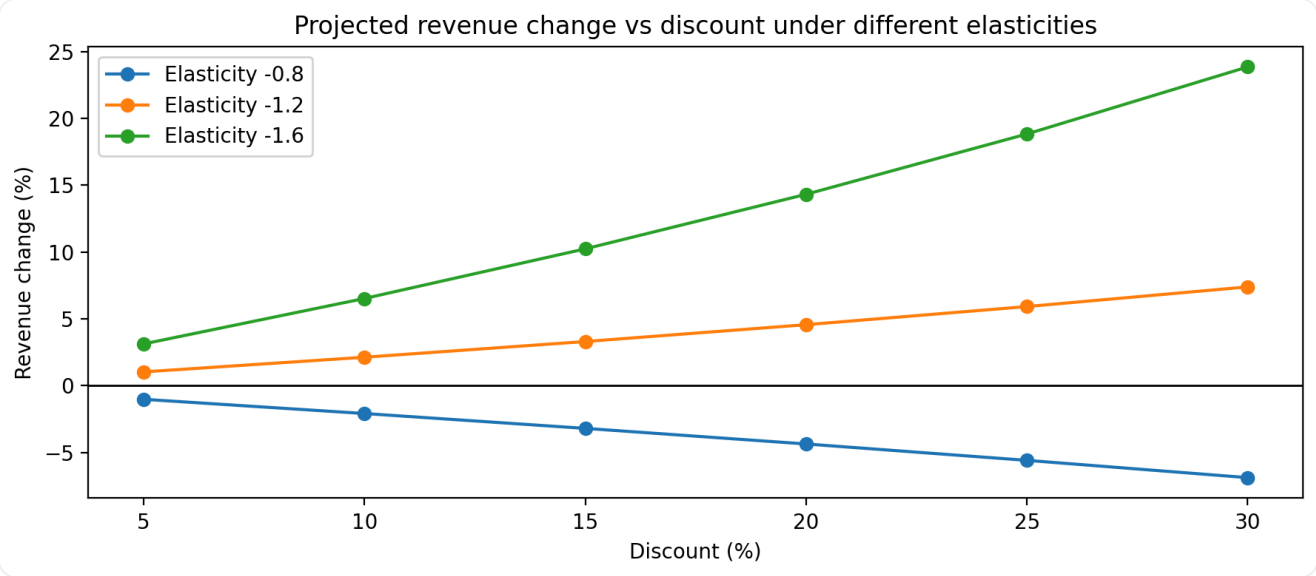
discount	elasticity	qty_change_pct	revenue_change_pct	projected_revenu
5%	-1.6	8.6	3.1	6259910

discount	elasticity	qty_change_pct	revenue_change_pct	projected_revenu
10%	-1.6	18.4	6.5	6466314
15%	-1.6	29.7	10.2	6691923
20%	-1.6	42.9	14.3	6939821
25%	-1.6	58.5	18.8	7213824
30%	-1.6	76.9	23.9	7518712
5%	-1.2	6.3	1.0	6132783
10%	-1.2	13.5	2.1	6199459
15%	-1.2	21.5	3.3	6270736
20%	-1.2	30.7	4.6	6347231
25%	-1.2	41.2	5.9	6429690
30%	-1.2	53.4	7.4	6519025
5%	-0.8	4.2	-1.0	6008236
10%	-0.8	8.8	-2.1	5943617
15%	-0.8	13.9	-3.2	5876058
20%	-0.8	19.5	-4.4	5805241
25%	-0.8	25.9	-5.6	5730791
30%	-0.8	33.0	-6.9	5652257

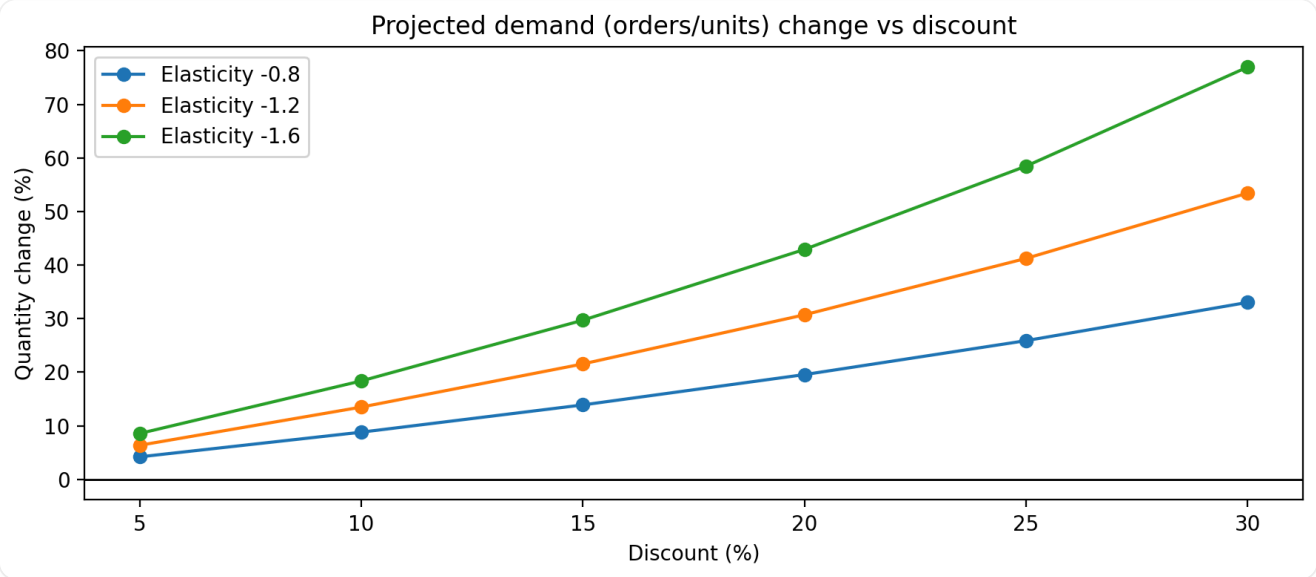
A few examples stand out:

- At **10% discount**, revenue ranges from **-2.1%** to **+6.5%** depending on how strongly demand responds.
- At **20% discount**, revenue could fall **4.4%** in a weak-response case or rise **14.3%** in a strong-response case.

Download



Download

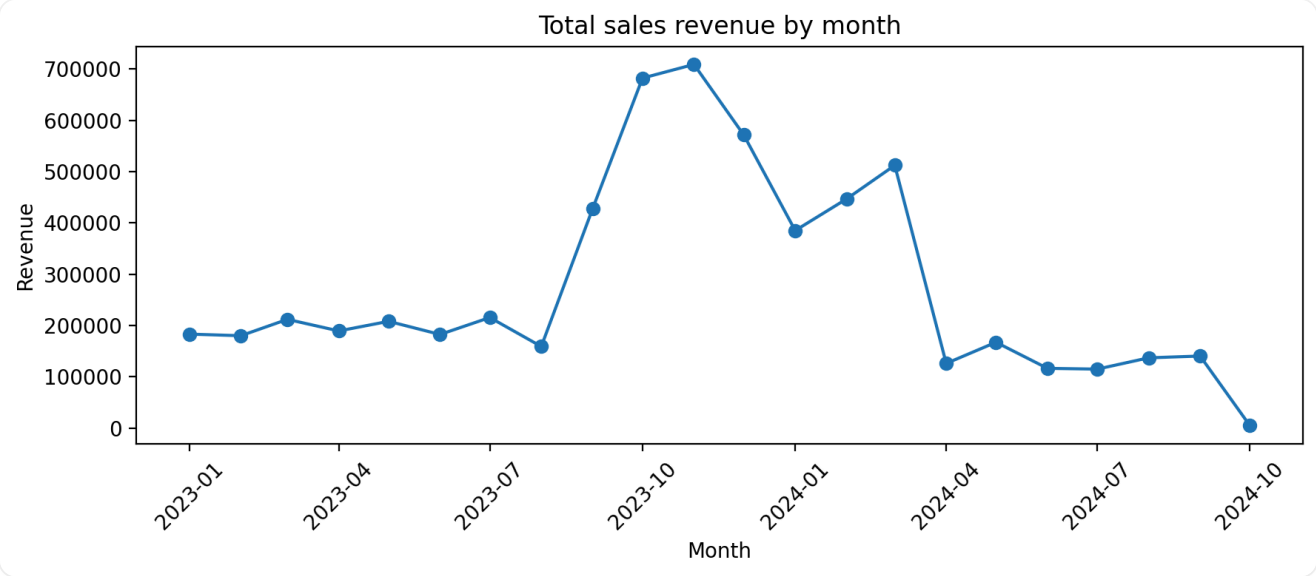


These charts make the tradeoff visible: promotions lower the amount earned per order, so they only pay off when they trigger enough extra buying.

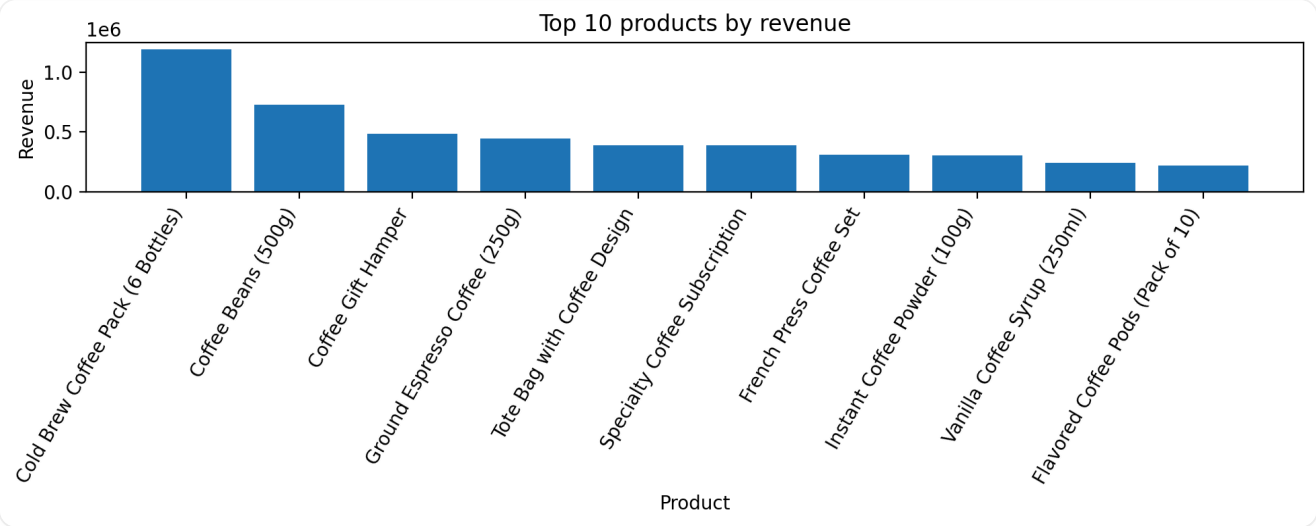
Sales are spread across time and strong products

The broader sales trend and product concentration help round out the picture. Revenue changes over time rather than staying flat, and a relatively small group of products contributes a large share of sales.

Download



Download



Together, these views support the same overall narrative: revenue growth depends on a mix of **where customers buy**, **how often they return**, and **how strongly they respond to pricing**.

Closing summary

- **Pune is the top-performing city** with revenue of **1,258,290**.
- Its lead is driven mainly by **higher order frequency**, not higher order value.
- Pune's customers account for about **10.5% of customers**, but roughly **20.6% of orders** and **20.7% of revenue**.
- Its strongest products are the same products that perform well overall, pointing to **repeat demand**, not a one-time product effect.
- Promotions can increase revenue, but only when the extra demand is strong enough to offset the lower selling price.

The strongest takeaway from the analysis is that **customer buying frequency is the biggest lever behind city performance**, and any promotion strategy must be judged by whether it creates enough additional demand to protect or grow revenue.